

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 7 INTEGRATION & ISSUE LOG

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme:	AnalystLab Africa Experience Lab
Workstream:	Project Management
Phase:	Week 7 — Testing, Refinement & End-to-End Validation
Version:	2.0
Status:	Controlled Working Log
Log Owner:	Project Management
Control Basis:	Week 7 testing evidence, approved outputs, cross-track dependencies, integration findings and validation results

1. PURPOSE

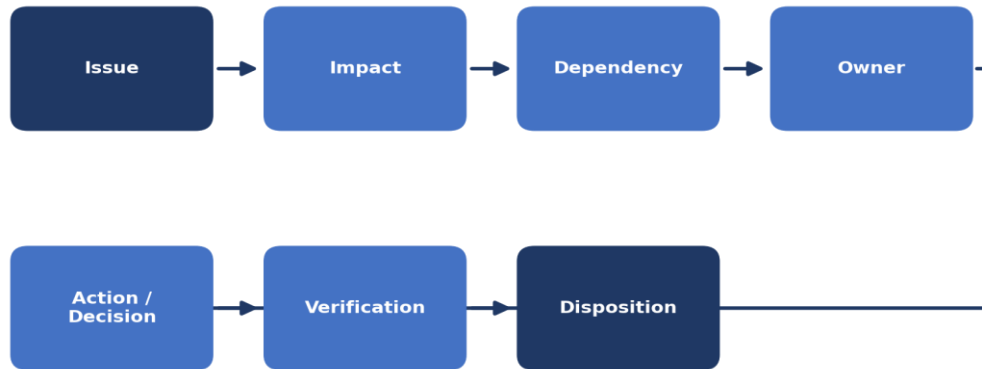
The Integration & Issue Log provides the controlled record of material integration issues, interface gaps and cross-track dependencies identified during Week 7 testing and validation.

Its purpose is to maintain traceability from issue identification through assessment, corrective action, verification and disposition, with particular focus on whether outputs produced by one workstream are suitable for use by another.

The log concentrates on issues that may affect:

- Cross-track handoffs and interface compatibility;
- Acceptance of upstream outputs;
- Operational interpretation of analytical or model outputs;
- GenAI safety, grounding, consistency and availability;
- End-to-end solution validation; and
- Week 8 integration and readiness.

Control Sequence



2. ISSUE STATUS CONVENTION

Status	Control Meaning
Open	Issue identified and requiring action; resolution has not yet been demonstrated.
In Progress	Corrective or investigative action is actively underway.
Blocked	Progress cannot continue because a required dependency, environment, decision or external condition is unavailable.
Pending Verification	Corrective action has been completed but objective verification or re-test remains outstanding.
Resolved	Corrective action has addressed the identified condition and evidence supports resolution, subject to any required downstream confirmation.
Accepted Residual	Limitation remains but has been explicitly assessed and accepted with documented ownership and control.
Closed	Resolution or disposition has been verified, downstream impact assessed, and no further action is required within the current project scope.

Severity

Severity	Definition
Critical	Prevents or materially threatens a critical integration, safety boundary, E2E validation or readiness gate.
High	Materially affects acceptance, dependency completion, validation confidence or downstream use.
Medium	Requires controlled resolution but does not currently prevent a critical project path.
Low	Limited impact with no material effect on current integration or readiness.

Severity describes impact; priority determines the urgency of response.

3. MASTER INTEGRATION & ISSUE LOG

Each issue record below captures the interface, condition, impact, dependency, ownership, required action and current status.

IIL-001 (DS → ML) — Final RF candidate	
Severity / Priority	Critical / Critical
Issue / Current Condition	Final DS-selected candidate has not yet been transferred and integrated into the ML environment.
Impact	Actual candidate compatibility and execution remain unverified.
Dependency / Blocker	Controlled DS→ML handoff
Owner	DS / ML
Required Action / Decision	Transfer candidate, confirm receipt, configuration and successful execution.
Target	Before final integration
Evidence Required	Handoff record + successful execution evidence
Status	Open

IIL-002 (DS → ML) — 33- feature schema / preprocessing	
Severity / Priority	Critical / Critical
Issue / Current Condition	Compatibility of the refined 33-feature structure and preprocessing requirements has not been formally reconciled in the ML environment.
Impact	Risk of feature-order, transformation or input mismatch during scoring.
Dependency / Blocker	IIL-001
Owner	DS / ML
Required Action / Decision	Reconcile feature structure, ordering, preprocessing and model input contract.
Target	Before candidate execution
Evidence Required	Schema reconciliation + scoring evidence
Status	Open

IIL-003 (DS → PM) — Classification threshold	
Severity / Priority	Critical / Critical

IIL-003 (DS → PM) — Classification threshold	
Issue / Current Condition	The candidate threshold has not been formally accepted as the operational decision rule.
Impact	Model output cannot be translated into an approved operational classification with a controlled acceptance basis.
Dependency / Blocker	Performance evidence + operational trade-off
Owner	DS / PM
Required Action / Decision	Assess threshold trade-off and document acceptance basis and configuration.
Target	Before E2E validation
Evidence Required	Decision record + configured threshold
Status	Open

IIL-004 (ML → Decision → Intervention) — Score-to- action workflow	
Severity / Priority	Critical / Critical
Issue / Current Condition	A validated pathway from model probability to risk classification, operational decision and intervention has not yet been established.
Impact	Technical scoring cannot yet be demonstrated as a complete HealthConnect intervention workflow.
Dependency / Blocker	IIL-003 + final model
Owner	ML / PM
Required Action / Decision	Define decision logic, risk classification and intervention pathway; validate end-to-end.
Target	Before Week 8
Evidence Required	E2E score → decision → intervention evidence
Status	Open

IIL-005 (Analytics → PM) — Power BI KPI layer	
Severity / Priority	High / High
Issue / Current Condition	Formal reconciliation of dashboard KPI calculations remains outstanding.
Impact	Dashboard acceptance remains constrained because reported measures have not been fully reconciled against the analytical basis.
Dependency / Blocker	Dataset + dashboard access

IIL-005 (Analytics → PM) — Power BI KPI layer	
Owner	Analytics
Required Action / Decision	Reconcile KPI definitions and calculations against the approved analytical dataset.
Target	Week 7 closure
Evidence Required	Reconciliation evidence
Status	Open

IIL-006 (Analytics → PM) — Dashboard filters / interactions	
Severity / Priority	High / High
Issue / Current Condition	Runtime validation of dashboard filters and interactions has not been fully evidenced.
Impact	Segmented views cannot yet be treated as fully validated for decision use.
Dependency / Blocker	Dashboard runtime environment
Owner	Analytics
Required Action / Decision	Execute structured filter and interaction checks and record outcomes.
Target	Week 7 closure
Evidence Required	Screenshots / exports + test results
Status	Open

IIL-007 (Analytics → DS) — EDA findings / model inputs	
Severity / Priority	High / High
Issue / Current Condition	Analytical findings are available, but downstream assessment, incorporation or formal disposition by DS is not fully evidenced.
Impact	Weakens traceability between analytical evidence and model-development decisions.
Dependency / Blocker	DS acknowledgement and assessment
Owner	Analytics / DS
Required Action / Decision	Confirm receipt, assessment and incorporation or documented disposition of relevant findings.
Target	Before model acceptance
Evidence Required	Handoff + disposition evidence
Status	Open

IIL-008 (GenAI → Test Environment) — Scenario execution	
Severity / Priority	High / High
Issue / Current Condition	Two planned GenAI scenarios could not be executed because of Gemini 503/high-demand availability issues.
Impact	Required evidence for those scenarios remains unavailable; they cannot be classified as pass or fail.
Dependency / Blocker	External service availability
Owner	GenAI
Required Action / Decision	Re-execute blocked scenarios when service availability permits.
Target	Before Week 7 closure
Evidence Required	Execution transcript + result
Status	Blocked

IIL-009 (GenAI → HealthConnect) — Response consistency	
Severity / Priority	High / High
Issue / Current Condition	Broader consistency testing remains incomplete beyond the scenarios already executed.
Impact	Current results do not establish sufficient assurance across varied but equivalent user phrasing.
Dependency / Blocker	Expanded scenario set
Owner	GenAI
Required Action / Decision	Execute structured consistency scenarios and compare response behaviour.
Target	Before final validation
Evidence Required	Comparative scenario results
Status	Open

IIL-010 (GenAI → HealthConnect) — Privacy boundary	
Severity / Priority	Critical / Critical
Issue / Current Condition	Privacy-sensitive scenarios have not yet been executed.

IIL-010 (GenAI → HealthConnect) — Privacy boundary	
Impact	The intended privacy/safety boundary remains unvalidated under test conditions.
Dependency / Blocker	Approved scenarios + test environment
Owner	GenAI
Required Action / Decision	Execute privacy-sensitive scenarios and document disposition.
Target	Before E2E
Evidence Required	Scenario evidence + disposition
Status	Open

IIL-011 (GenAI → HealthConnect) — Emergency escalation	
Severity / Priority	High / High
Issue / Current Condition	Existing emergency tests were successful within tested scenarios, but broader indirect or vague emergency phrasing requires additional validation.
Impact	Current evidence does not establish complete coverage of emergency-related language variation.
Dependency / Blocker	Expanded scenario coverage
Owner	GenAI
Required Action / Decision	Extend emergency scenarios and perform regression testing.
Target	Before final validation
Evidence Required	Scenario + regression evidence
Status	Open

IIL-012 (DS → PM) — Model validation	
Severity / Priority	High / High
Issue / Current Condition	Refined RF shows a train/test ROC-AUC gap of approximately 0.0571, indicating a remaining generalisation limitation.
Impact	Limitation is relevant to model acceptance and interpretation of expected performance.
Dependency / Blocker	Model acceptance criteria
Owner	DS

IIL-012 (DS → PM) — Model validation	
Required Action / Decision	Document limitation, assess materiality and carry any required control into acceptance/monitoring.
Target	Before model acceptance
Evidence Required	Validation evidence + disposition
Status	Open

IIL-013 (DS → PM) — Segment performance	
Severity / Priority	Medium / Medium
Issue / Current Condition	Weaker precision on Sunday and weaker recall for Specialist Consultation were observed during segment analysis.
Impact	Segment-specific performance may affect operational interpretation if treated without appropriate context.
Dependency / Blocker	Further assessment / corroboration
Owner	DS
Required Action / Decision	Assess and document the operational significance and limitations of the segment findings.
Target	Before final acceptance
Evidence Required	Segment analysis + disposition
Status	Open

IIL-014 (ML → Integrated Solution) — ML scoring pipeline	
Severity / Priority	High / High
Issue / Current Condition	Technical ML validation is substantially demonstrated, but operational integration into the complete HealthConnect workflow remains unvalidated.
Impact	Technical pipeline completion could otherwise be mistaken for solution readiness.
Dependency / Blocker	IIL-003, IIL-004, IIL-015
Owner	ML / PM
Required Action / Decision	Maintain separate technical and operational acceptance; complete downstream workflow validation.
Target	Before Week 8 readiness
Evidence Required	E2E validation evidence

IIL-014 (ML → Integrated Solution) — ML scoring pipeline	
Status	Open

IIL-015 (All Tracks → Integrated Solution) — HealthConnect E2E workflow	
Severity / Priority	Critical / Critical
Issue / Current Condition	The complete cross-track workflow has not yet been validated end-to-end.
Impact	There is no objective evidence yet that the independently tested components operate coherently as one solution.
Dependency / Blocker	Critical upstream interfaces
Owner	PM / All Tracks
Required Action / Decision	Execute E2E validation once critical dependencies are cleared or appropriately dispositioned.
Target	Before Week 8 integration
Evidence Required	E2E test evidence + disposition
Status	Open

IIL-016 (Testing → PM) — Risk / dependency controls	
Severity / Priority	High / High
Issue / Current Condition	Week 7 findings require reconciliation with the current Risk & Dependency Register.
Impact	Project risk position may not fully reflect the latest testing and integration evidence.
Dependency / Blocker	Active Week 7 findings
Owner	PM
Required Action / Decision	Review and update risk/dependency status and link material issues to the appropriate controls.
Target	Week 7 closure
Evidence Required	Updated register + traceability
Status	Open

IIL-017 (All Tracks → PM) — Week 8 readiness	
Severity / Priority	Critical / Critical
Issue / Current Condition	Critical integration and E2E conditions remain outstanding; readiness cannot yet be conclusively established.
Impact	Final integration readiness remains conditional on resolution or formal disposition of critical issues.
Dependency / Blocker	IIL-001–IIL-015 as applicable
Owner	PM / All Tracks
Required Action / Decision	Complete critical actions, verify evidence and conduct formal Week 8 readiness assessment.
Target	Week 8 gate
Evidence Required	Approved readiness assessment
Status	Open

4. CROSS-TRACK INTEGRATION CONTROL MATRIX

Interface	Integration Purpose	Acceptance Condition	Current Position
Analytics ↔ DS	Transfer analytical findings and validated data evidence into model development.	Consistent definitions, usable analytical evidence and traceable downstream disposition.	Partially evidenced
DS ↔ ML	Transfer and operationalise the selected model candidate.	Candidate, feature schema, preprocessing and configuration accepted and successfully executed.	Outstanding
ML ↔ Operational Layer	Convert model output into an operationally usable decision.	Valid score, accepted threshold and defined decision logic.	Outstanding
Decision ↔ Intervention	Connect risk classification to an actionable response.	Defined and testable intervention pathway linked to the decision output.	Outstanding
ML ↔ GenAI	Establish any actual technical interface where model output is consumed by GenAI.	Explicit interface, data contract and successful technical validation.	No direct technical integration evidenced
GenAI ↔ HealthConnect	Validate safe, grounded and operationally appropriate conversational support.	KB grounding, emergency escalation, consistency and privacy boundaries adequately tested.	Partially validated

Interface	Integration Purpose	Acceptance Condition	Current Position
All Tracks ↔ Integrated Solution	Demonstrate that validated components operate coherently as one HealthConnect solution.	E2E workflow successfully executed and accepted.	Not demonstrated

5. INTEGRATION ISSUE CATEGORIES

The following categories define the control areas used when assessing new or emerging integration issues. They are not assertions that every listed condition currently exists.

A. Data & Definition Integrity

Covers:

- Schema changes or incompatibilities;
- Inconsistent field or metric definitions;
- Missing or incomplete historical information;
- Feature/target inconsistencies;
- Downstream impact of analytical changes.

B. Model & Pipeline Integration

Covers:

- Candidate-model handoff;
- Feature and preprocessing compatibility;
- Model input/output contract;
- Reproducibility;
- Execution compatibility;
- Scoring behaviour.

C. Operational Integration

Covers:

- Threshold acceptance;
- Score interpretation;
- Risk classification;
- Decision rules;
- Intervention mapping;
- Workflow usability and ownership.

D. GenAI Integration & Reliability

Covers:

- Service availability;
- Scenario execution;
- Response consistency;

- Knowledge Base grounding;
- Emergency escalation;
- Unsupported information;
- Privacy-sensitive interactions.

E. Cross-Track Governance

Covers:

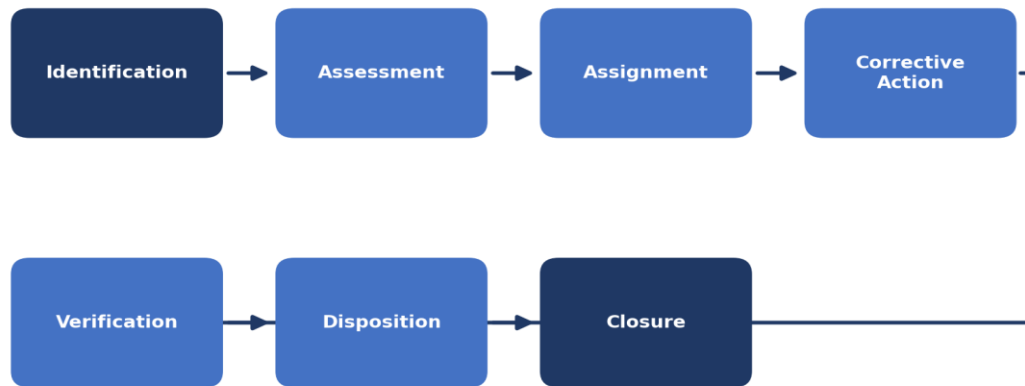
- Unresolved interface dependencies;
- Incomplete handoff evidence;
- Acceptance decisions;
- Safety/privacy boundaries;
- Unresolved issues affecting project gates.

6. DEPENDENCY & BLOCKER CONTROL

ID	Dependency / Blocker	Effect on Integration	Current Control
DEP-I01	Final DS candidate → ML integration	Blocks validation of the actual selected model within the ML pipeline.	Controlled DS→ML handoff required.
DEP-I02	33-feature schema → ML preprocessing contract	Blocks confirmation that candidate inputs are transformed consistently.	Schema and preprocessing reconciliation required.
DEP-I03	Model performance evidence → operational threshold	Blocks formal acceptance of the decision rule.	Threshold assessment and decision record required.
DEP-I04	Accepted threshold → decision/intervention pathway	Blocks score-to-action validation.	Define and test operational workflow.
DEP-I05	Dashboard evidence → analytical acceptance	Limits acceptance of current dashboard outputs.	KPI and interaction validation required.
DEP-I06	Analytics findings → DS disposition	Limits traceability of analytical evidence into model decisions.	Confirm downstream assessment/disposition.
DEP-I07	GenAI service availability → remaining scenarios	Blocks execution of two planned scenarios.	Re-test when service becomes available.
DEP-I08	Accepted interfaces → E2E workflow	Blocks integrated solution validation.	Resolve critical interfaces before E2E execution.

7. ISSUE LIFECYCLE & CLOSURE CONTROL

All material issues follow the controlled lifecycle:



An issue may only be Closed where:

1. the required corrective action has been implemented;
2. the affected condition has been re-tested or objectively verified;
3. the expected condition has been demonstrated, or an explicit residual limitation has been accepted;
4. downstream impact has been assessed; and
5. verification evidence has been recorded.

Accepted Residual is not equivalent to resolution. It may only be used where the remaining limitation is documented, assigned to an owner, assessed for impact and supported by an explicit acceptance basis.

8. PM ESCALATION CONTROL

Escalation is recorded against an issue where its impact exceeds routine workstream resolution and requires PM intervention to protect integration, validation, acceptance, or Week 8 readiness. For HealthConnect, escalation is required where an issue:

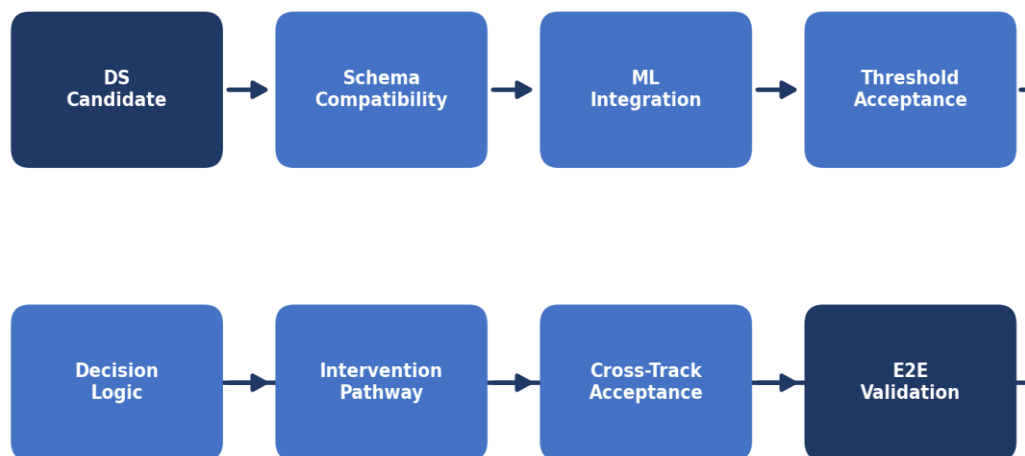
- Breaks or delays a critical cross-track handoff, particularly the DS candidate → ML integration → decision rule → intervention pathway;
- Prevents an output from being accepted as a valid input to the dependent workstream;
- Creates a material discrepancy in data definitions, feature structure, preprocessing, model configuration, KPI calculations, or downstream interpretation;
- Leaves a required operational acceptance decision unresolved, including model threshold, risk classification, or score-to-action logic;
- Affects GenAI safety, privacy, escalation behaviour, or Knowledge Base grounding, or prevents required scenarios from being adequately tested;
- Prevents objective verification of a component or interface and therefore blocks end-to-end HealthConnect validation;

- Depends on an external service or environment whose unavailability prevents required evidence from being generated or re-tested;
- Introduces a limitation that could materially alter a previously accepted assumption, requirement, dependency, or readiness position;
- Remains unresolved, unverified, or without an agreed disposition at a defined project gate or Week 8 integration checkpoint; or
- Changes the evidence basis on which solution validation or implementation readiness would otherwise be assessed.

Escalated issues must carry a clear impact, dependency, accountable owner, required decision or corrective action, target resolution point, and verification evidence. PM escalation does not replace workstream ownership; it establishes the coordination, decision, and gate-control mechanism required where resolution crosses workstream boundaries or affects project readiness.

9. CURRENT INTEGRATION POSITION

Week 7 has established substantial component-level validation, particularly across Data Science, ML Engineering and GenAI testing. However, the critical integration chain has not yet been fully demonstrated:



The principal control concern is therefore no longer whether individual workstreams have produced outputs, but whether those outputs have crossed their interfaces, been accepted by dependent workstreams, been re-tested where necessary, and demonstrated sufficient coherence to support final solution validation.

The Integration & Issue Log therefore remains open, with Week 8 readiness dependent on the resolution or formal disposition of the critical integration conditions identified above.